SMART INDIA HACKATHON 2026

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THREAD ZERO

- **Problem Statement ID – PS26198**
- **Problem Statement Title - Student Innovation**
- **Theme – Healthcare**
- **PS Category - Software**
- **Team ID – 150145**
- **Team Name – Thread Zero**



Thread Zero

MedClear

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THE PROBLEM

1. **Complex Medical Language**
   Medical reports contain technical terms that patients struggle to understand.

2. **Important Results Are Missed**
   Abnormal or critical values can be difficult for patients to identify.

3. **Language Barrier**
   Medical information is often unavailable in simple Indian-language explanations.

4. **Dependence on Others**
   Patients may need doctors or family members to interpret their reports.

HOW IT ADDRESSES THE PROBLEM

MedClear is an AI-powered platform that turns complex medical reports and prescriptions into simple, easy-to-understand language. It highlights important findings and supports multiple languages, helping patients and caregivers understand their health better and have clearer conversations with doctors.



Acts as a prerequisite to doctors appointments and enables clearer, more informed patient–doctor conversations.



Empowers caregivers with a clear understanding of patients' medical reports and prescriptions.



Bridges the gap between medical background and personal health understanding, reducing confusion and panic.



Keeps families informed through timely alerts, ensuring everyone stays aware of the patient's health situation.



Reduces unnecessary misdirected refrals and financial costs while saving time, effort, and unnecessary hassle.

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TECHNICAL APPROACH

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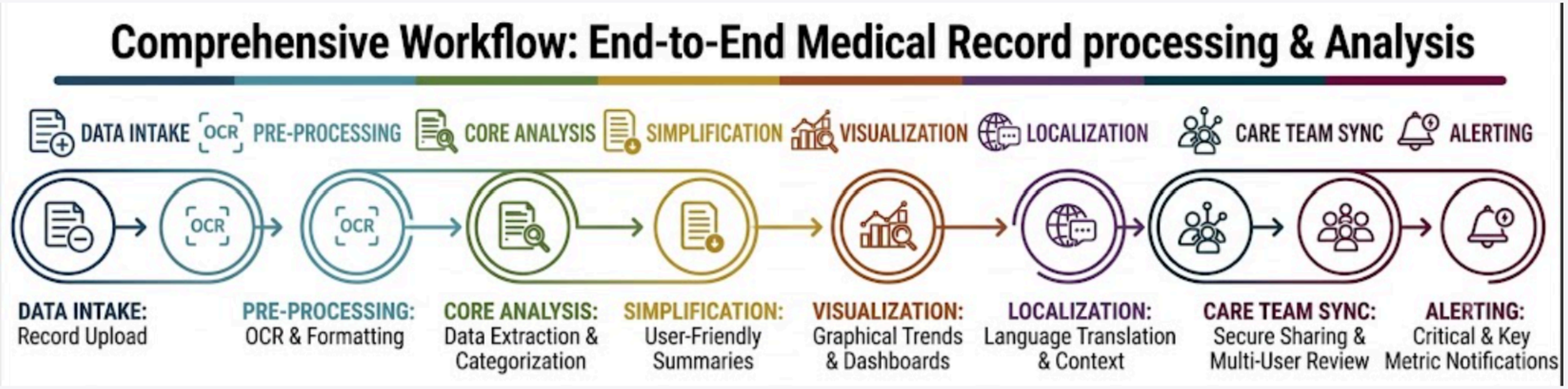
TECH STACK

Ingestion & OCR OpenCV, PyMuPDF, Tesseract OCR	**Regional Localization** IndicTrans2, Bhashini APIs
Clinical Extraction (NLP) scispaCy, spaCy, Med7	**Backend & Persistence** FastAPI, Celery, PostgreSQL
Simplification & Validation Llama 3, BioMistral, TextStat	**Frontend & UI** React.js / Next.js, Tailwind CSS

CASE STUDY

A patient receives a complex blood-test report and prescription containing unfamiliar medical terms and multiple values.

- Before MedClear: The patient and caregiver struggle to understand abnormal values, medical terminology, and prescribed instructions, leading to confusion and dependence on others.
- With MedClear: The report is converted into a simple summary, important values are visually highlighted, prescriptions are explained clearly, and the summary can be shared with family members in a regional language.
- Outcome: Better health understanding, reduced confusion, informed doctor conversations, and improved caregiver coordination.



PROTOTYPE

https://youtu.be/WwitLx2RKnE

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FEASIBILITY AND VIABILITY

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CHALLENGES AND MITIGATIONS

1. **Medical jargon is hard for users to understand.**
   Mitigation: AI-based text simplification and term mapping convert complex terms into simpler explanations.

2. **Uploaded reports can be in various formats (PDF/images) affecting accuracy.**
   Mitigation: Document processing tools and OCR reliably extract text across file types.

3. **Simplification can risk altering clinical meaning.**
   Mitigation: The system retains original medical context while simplifying language without changing key details.

VIABILITY

- Many patients struggle to understand complex medical reports written in technical language.
- Simplifying medical information improves health literacy and patient awareness.
- Additionally, features like multilingual support and visual explanations make the system useful for a large and diverse population.

FEASIBILITY

- Technically Feasible: Uses proven technologies such as OCR, NLP, LLMs, clinical databases, readability scoring, and translation APIs to process medical reports end-to-end.
- Operationally & Financially Feasible: A simple upload → analyse → simplify → highlight workflow can be deployed using existing cloud services and scaled based on user demand.
- Scalable & Safe: Modular architecture supports more languages, mobile apps and hospital integration, while clinical grounding

SCALABILITY

- The platform can be expanded to support more languages, additional medical databases, integration with hospital systems, and mobile applications.
- With proper cloud deployment and optimized AI models, the system can handle increasing data and user requests efficiently.

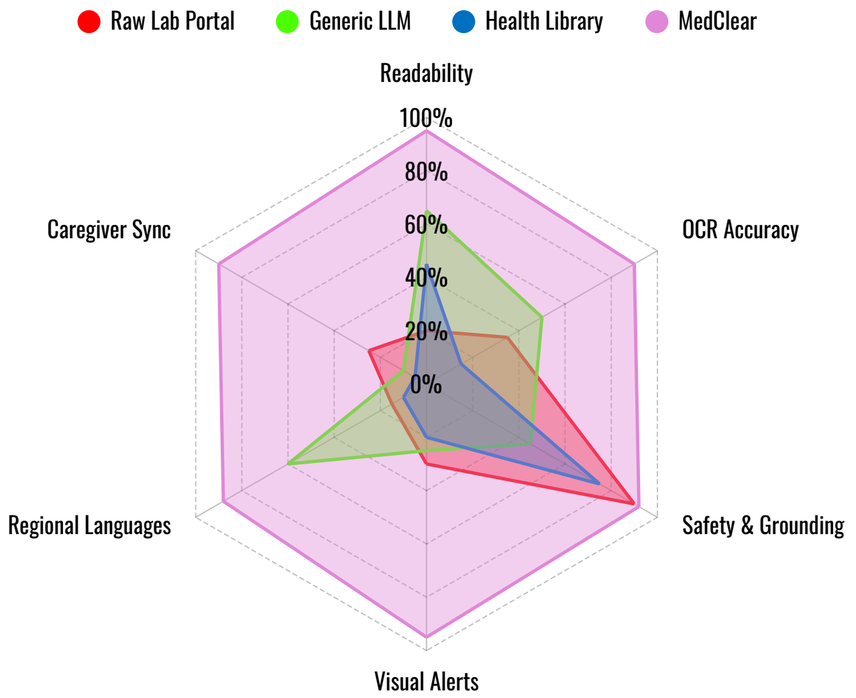
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IMPACT AND BENEFITS

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IMPACT

- **Boosts Health Literacy:** Simplifies clinical jargon to an 8th-grade reading level for clear patient understanding.
- **Bridges Language Barriers:** Delivers accurate medical summaries in regional Indian languages.
- **Speeds Up Care:** Enables real-time caregiver sync and visual alerts for abnormal lab values.
- **Ensures Patient Safety:** Grounds simplifications in clinical databases to prevent hallucinations and preserve context.



BENEFITS

- **Full Report Analysis:** Explains the entire report context rather than isolated terms.
- **Multilingual Access:** Translates complex findings into regional Indian languages.
- **Family Sync:** Lets users share health status directly with family or caregivers.
- **Critical Alerts**: Detects abnormal results to prompt timely medical action.

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RESEARCH AND REFERENCES

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Research Papers

1. Improving Readability in Medical Reports Using Artificial Intelligence: A Comprehensive Review
Link: https://ijmraas.misj.net/index.php/ijmraas/article/view/39

2. Machine Learning in Medicine: A Practical Introduction to Natural Language Processing
Link: https://bmcmedresmethodol.biomedcentral.com/articles/10.1186/s12874021-01347-1

3. Assessing AI Simplification of Medical Texts: Readability and Content Fidelity
Link: https://pubmed.ncbi.nlm.nih.gov/39667051

4. Can Artificial Intelligence Improve Patient Educational Material Readability?
Link: https://pubmed.ncbi.nlm.nih.gov/39720869

5. Simplification and Translation of Medical Reports Using Large Language Models
Link: https://www.preprints.org/manuscript/202508.0955/v1

6. AI in Healthcare: Simplifying Medical Reports for Enhanced Patient Comprehension
Link: https://ijirt.org/publishedpaper/IJIRT175571_PAPER.pd

Databases & Datasets

1. MIMIC-III Clinical Database
Link: https://physionet.org/content/mimic i/1.4/ A large dataset containing de-identified clinical records of over 40,000 ICU patients, including lab results, clinical notes, and patient data.

2. MIMIC-IV Clinical Database
Link: https://physionet.org/content/mimiciv/3.1/ An updated version of MIMIC- I that provides large-scale hospital and ICU medical data for healthcare AI research.

3. i2b2 Clinical NLP Dataset
Link: https://www.i2b2.org/NLP/DataSets/ A dataset containing de-identified clinical notes used for research in medical Natural Language Processing (NLP).

4.MIMIC-IV-Note
Clinical NotesDe-identified free-text clinical notes.
Best suited for NLP, medical text understanding and report summarization.
https://physionet.org/content/mimic-iv-note/

5.MIMIC-CXR
Medical Reports + X-raysContains free-text radiology reports paired with chest X-rays.
Very relevant to MedClear's report-analysis pipeline.
https://physionet.org/content/mimic-cxr/2.1.0/